



examGenome CheatSheet on Algebra

Sl.no	Content
1	What is algebra? Answer: Algebra is a part of math where letters (like x or y) are used to stand for unknown numbers. Explanation: Instead of only working with numbers like $2 + 3$, algebra uses symbols to represent numbers we don't know yet—like $x + 3 = 7$.
2	What is a variable? Answer: A variable is a letter or symbol that stands for a number that can change or is unknown. Explanation: For example, in ' $x + 5 = 10$ ', x is a variable. It could be any number, but here it must be 5 to make the equation true.
3	What is an expression in algebra? Answer: An expression is a group of numbers, variables, and operations (like $+$, $-$, $\times$) without an equals sign. Explanation: Examples: $3x$, $x + 4$, $2y - 5$. Expressions don't say two things are equal—they just show a calculation.
4	What is an equation? Answer: An equation is a statement that two expressions are equal, using an equals sign ($=$). Explanation: Example: $2x + 1 = 7$. Equations can be solved to find the value of the variable.
5	What does it mean to 'solve' an equation? Answer: To find the value of the variable that makes the equation true. Explanation: For $x + 2 = 5$, solving means finding that $x = 3$, because $3 + 2 = 5$.
6	What is the rule for keeping an equation balanced? Answer: Whatever you do to one side of the equation, you must do the same to the other side. Explanation: If you add 3 to the left side, you must add 3 to the right side—so the equation stays true.
7	Can letters in algebra stand for any number? Answer: Yes! Letters can represent whole numbers, fractions, or decimals—it just depends on the problem. Explanation: For example, x could be 4, 0.5, or 100—whatever makes the equation or expression work.
8	What is a coefficient? Answer: A coefficient is the number multiplied by a variable. Explanation: In $5x$, the 5 is the coefficient. It means 5 times x . If you see just ' x ', the coefficient is 1 (not written).
9	What is a constant in algebra? Answer: A constant is a number that does not change and has no variable attached. Explanation: In the expression $4x + 7$, the 7 is a constant. It stays the same no matter what x is.

<i>Sl.no</i>	<i>Content</i>
10	Do algebraic expressions have units? Answer: Only if the problem involves real things like meters or dollars—but the letters themselves have no units. Explanation: If x stands for 'number of apples', then x has no unit, but '$5x$ apples' would be in apples. Algebra is mostly about numbers and rules, not units.

